RESULTS: KL REGULARIZATION AGAINST AN EMA REFERENCE

BRIAN BARTOLDSON

1. Setup

We train Qwen3-4B-Instruct-2507 with TBA on Countdown using asynchronous rollouts ($\Delta = 10$), $\alpha = 0.9$ for the EMA reference, $\beta = 0.005$, lr $= 10^{-6}$, $K = 16$ samples per problem, sequence length 4096, 720 training steps, evaluation every 20 steps.

Per-token KL.. Two choices for the per-token KL contribution to the advantage:

- *Exact KL:* $\log T_n(y) - \log E_n(y)$, where E_n is the EMA reference policy.
- *Approx KL:* $\frac{\alpha}{\Delta(1-\alpha)}\big(\log T_n(y) - \log T_{n-\Delta}(y)\big)$, the first-order surrogate derived in `main.tex` (eq. 2.10). This avoids a forward pass through E_n at training time.

Centering. For each rollout i, the per-token KL is summed over masked tokens to give a per-rollout KL term $K^{(i)} = \sum_t \text{kl\_per\_token}(y_t^{(i)})$. The advantage is then adjusted by

$$\text{advantage}^{(i)} \;\mathrel{+}= \; -\beta \cdot \big(K^{(i)} - \bar{K}\big), \tag{1}$$

where $\bar{K}$ is a centering term. Two choices:

- *K-centered:* $\bar{K} = \frac{1}{K}\sum_{j=1}^{K} K^{(j)}$, the within-group mean of $\{K^{(j)}\}_{j=1}^{K}$ over the K rollouts sharing a problem.
- *No centering:* $\bar{K} \equiv 0$, so advantage$^{(i)}$ $+= -\,\beta\, K^{(i)}$ directly.

2. 2×2 matrix: EMA reference

Crossing the two per-token KL choices with the two centering choices yields four cells:

Cell	kl_per_token(y)	Centering term $\bar{K}$
Exact, K-centered	$\log T_n(y) - \log E_n(y)$	$\frac{1}{K}\sum_{j=1}^{K} K^{(j)}$
Exact, no centering	$\log T_n(y) - \log E_n(y)$	0
Approx, K-centered	$\frac{\alpha}{\Delta(1-\alpha)}\big(\log T_n(y) - \log T_{n-\Delta}(y)\big)$	$\frac{1}{K}\sum_{j=1}^{K} K^{(j)}$
Approx, no centering	$\frac{\alpha}{\Delta(1-\alpha)}\big(\log T_n(y) - \log T_{n-\Delta}(y)\big)$	0

(In the K-centered rows, $K^{(j)}$ is built from the same row's kl_per_token.)

Cell	Peak eval (step)	Final eval
Exact, K-centered	0.789 (240)	0.656
Exact, no centering	**0.827** (620)	0.821
Approx, K-centered	0.794 (380)	0.548
Approx, no centering	**0.831** (520)	0.819

The headline observations:

1. Within either KL choice, removing the centering term strictly helps.
2. The approx-KL row tracks the exact-KL row closely: peaks differ by 0.005 and finals by 0.002 in the no-centering pair. This is the empirical justification for substituting the first-order surrogate (no extra forward) for the exact EMA KL.
3. The K-centered cells exhibit the same kind of late-run instability under both per-token KL choices, suggesting the failure mode is centering-related rather than KL-formula-related.

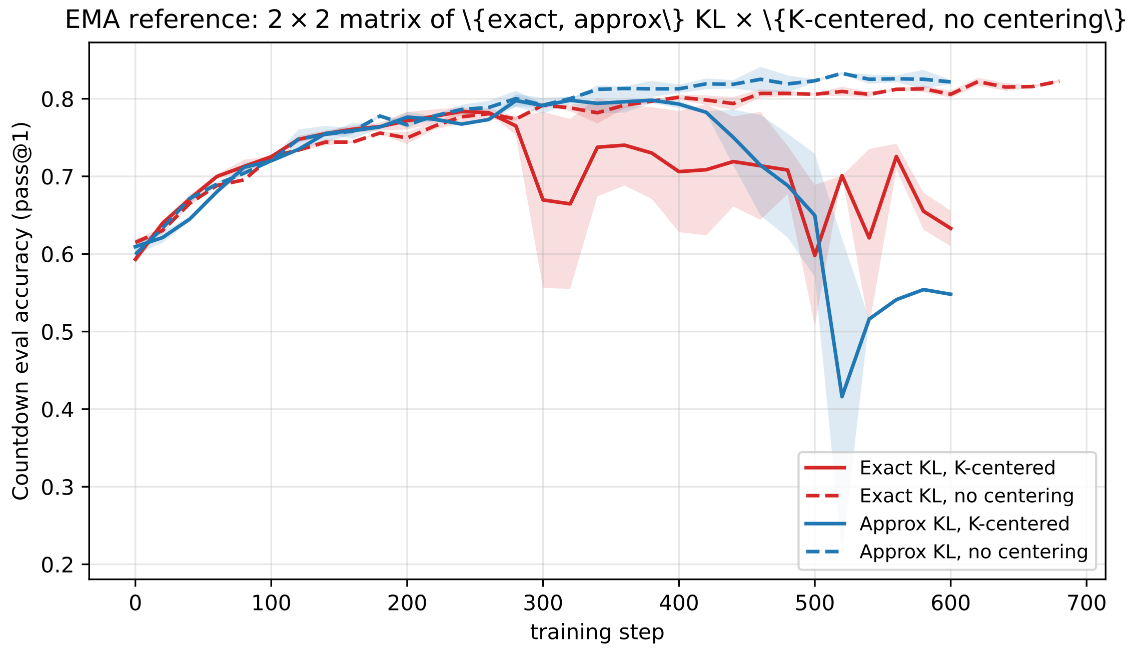

FIGURE 1. Eval accuracy on Countdown for the four cells of the EMA-reference matrix. Red: exact KL. Blue: approx KL. Solid: K-centered. Dashed: no centering. The two no-centering cells (dashed) plateau above 0.80 and peak above 0.82; the two K-centered cells (solid) peak earlier and degrade after step ≈ 300.

3. Reset reference (baseline)

Before the EMA work, the same form of objective was tested with a periodic-reset reference policy R (hard copy from the training policy every 50 rollout steps). The advantage update (1) is unchanged, but per-token KL is computed against R instead of E_n, and the source for $\log T(\cdot)$ becomes configurable. Let T_n denote the live training-policy forward at the current step and $T_{n-\Delta}$ the inference snapshot used during rollout generation. The per-sample per-token KL is

$$\log T_{\mathrm{src}_1}(y) - \log R(y), \tag{2}$$

and the K-group mean term substitutes a (potentially different) source T_{src_2}:

$$\bar{K} \;=\; \frac{1}{K} \sum_{i=1}^{K} \sum_{t} \Big(\log T_{\mathrm{src}_2}(y_t^{(i)}) - \log R(y_t^{(i)})\Big), \tag{3}$$

where each $\mathrm{src}_j \in \{T_n, T_{n-\Delta}\}$. We test three combinations, each with importance sampling on or off, all with K-centering:

Variant (short)	Per-sample $\log T_{\mathrm{src}_1}$	K-group mean $\log T_{\mathrm{src}_2}$
T/I	$\log T_n$ (live train forward)	$\log T_{n-\Delta}$ (inference snapshot)
T/T	$\log T_n$	$\log T_n$
I/I	$\log T_{n-\Delta}$	$\log T_{n-\Delta}$

The first slot is the per-sample $\log T$; the second is the K-mean $\log T$. Historically these mapped to “klMeanInference”, “klMeanTrain”, and “klAllInf” respectively.

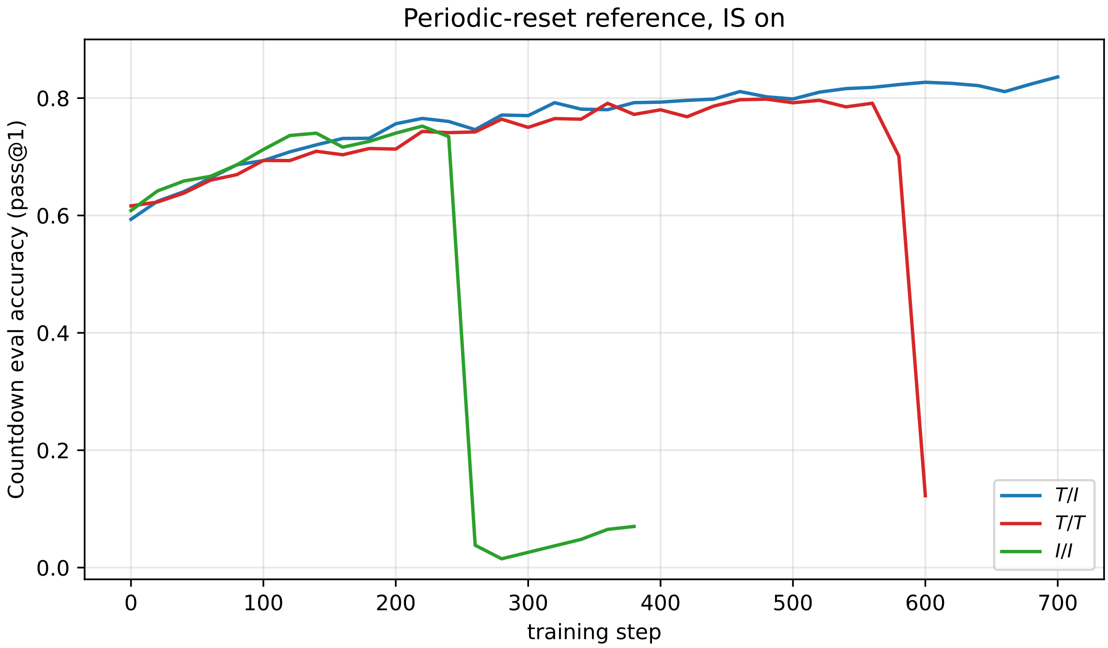

FIGURE 2. Eval accuracy on Countdown with periodic-reset reference and importance sampling on. T/T (red) collapses late in training; I/I (green) is unstable; only T/I (blue) trains stably to high accuracy.

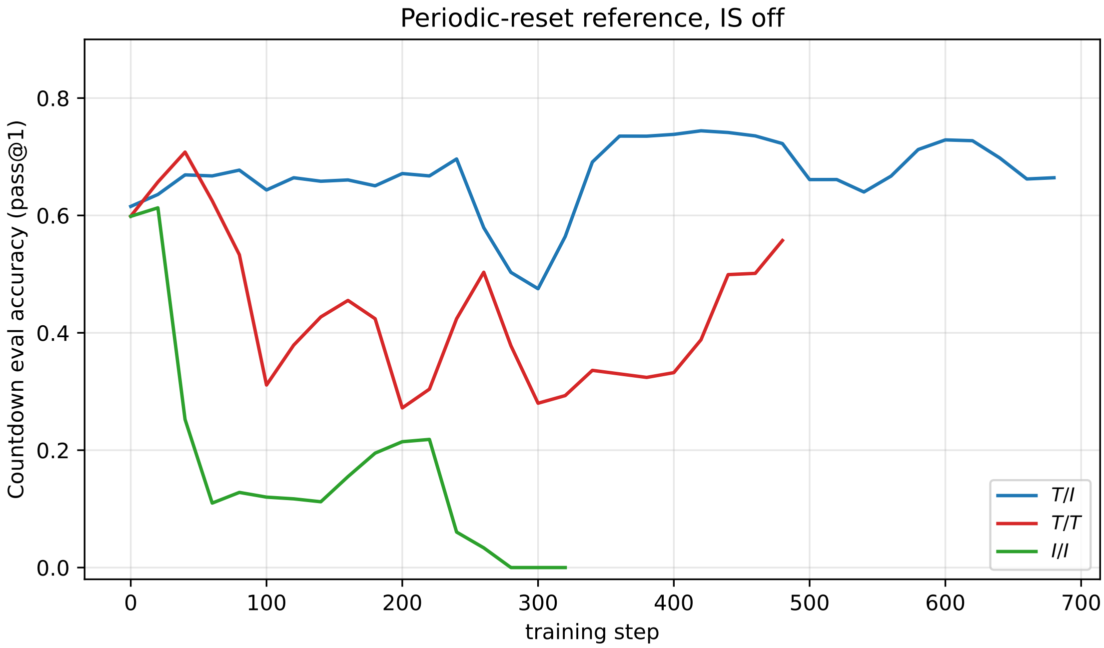

FIGURE 3. Eval accuracy on Countdown with periodic-reset reference and importance sampling off. The same ordering holds — T/I best, T/T collapses, I/I gets stuck near zero — and all three are weaker than their IS-on counterparts.

Run	Peak eval (step)	Final eval
T/I	0.836 (700)	0.836
T/T	0.798 (480)	0.122
I/I	0.752 (220)	0.070
T/I, no IS	0.744 (680)	0.664
T/T, no IS	0.708 (40)	0.557
I/I, no IS	0.613 (20)	0.000

The T/T collapse under reset reference mirrors the K-centered instability in the EMA matrix: in both cases, the per-sample and K-mean KL terms are computed from the *same* drifting source T_n, so the centering subtraction tracks the drift rather than canceling it.

4. Approximation accuracy

We track the per-token error of the first-order surrogate against the exact EMA KL inside `tba_loss`: `MAE` (mean absolute error over masked tokens) and `bias` (signed mean). Figure 4 compares the two no-centering cells (where the loss path differs but both have the same telemetry).

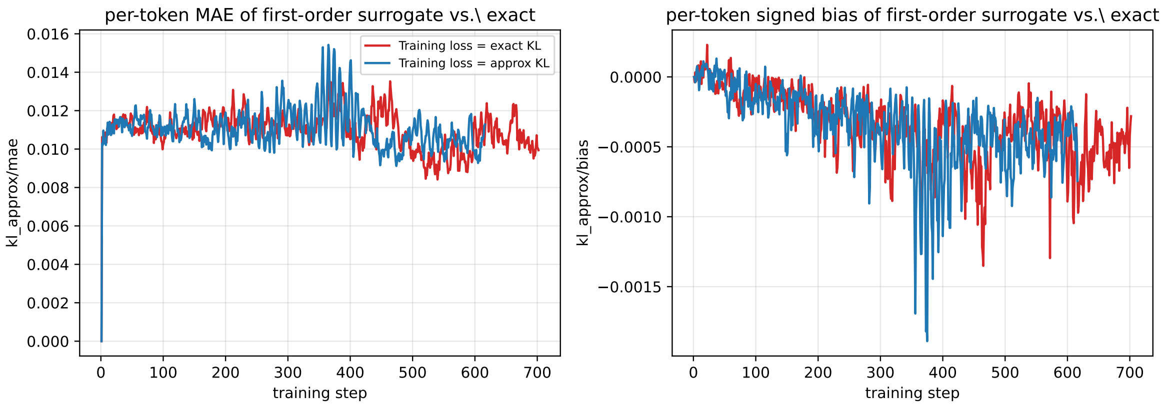

FIGURE 4. Per-token MAE (left) and signed bias (right) of the first-order surrogate $\frac{\alpha}{\Delta(1-\alpha)}\big(\log T_n(y) - \log T_{n-\Delta}(y)\big)$ against the exact $\log T_n(y) - \log E_n(y)$, averaged over masked tokens within each rollout. Both runs are no-centering EMA-reference cells; they differ only in which formula enters the loss (red: exact KL; blue: approx KL).

MAE stays in the low 10^{-2} band throughout training, and the signed bias is at most a few parts in 10^3. The approximation is empirically faithful, regardless of whether it is used in the loss.

5. Ablations

We probe three knobs in the no-centering / K-centered EMA cells. Some of the runs cited here are still in progress, so trend lines for the newest variants may be short.

5.1. β ablation on the K-centered cell (A).

Cell A (EMA exact + klMeanTrain centering) showed a late-run collapse around step ~ 300 at the baseline $\beta = 0.005$. Does scaling the KL penalty up rescue it?

5.2. Importance sampling ablation on no-centering.

The no-centering cells B and D (exact and approx) were the top-performing matrix entries at $\beta = 0.005$ with IS on. Turning IS off keeps the same conclusion that no centering is best, but at lower absolute accuracy and with late-run instability.

5.3. α ablation on no-centering exact.

Sweep the EMA decay $\alpha \in \{0.8, 0.9, 0.95\}$ while holding the effective regularization strength constant ($\beta \cdot c_\alpha = 0.0045$, where $c_\alpha = \alpha/(\Delta(1-\alpha))$).

6. Conclusion

Removing the K-group centering is the dominant lever: both EMA exact and EMA approx benefit. With centering off, the cheap first-order surrogate matches the exact EMA KL within 0.005 on peak eval and 0.002 on final eval, so it is a viable drop-in replacement that saves a forward pass through the reference model.

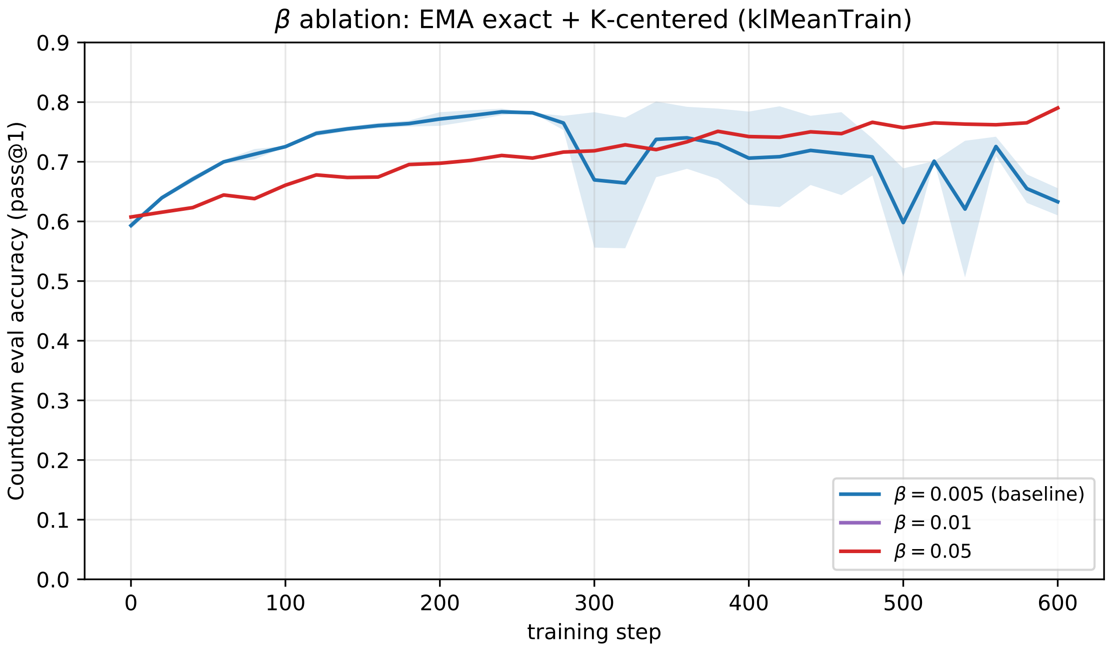

FIGURE 5. β ablation on K-centered exact KL (cell A). Larger β trades off learning speed against the K-centered instability.

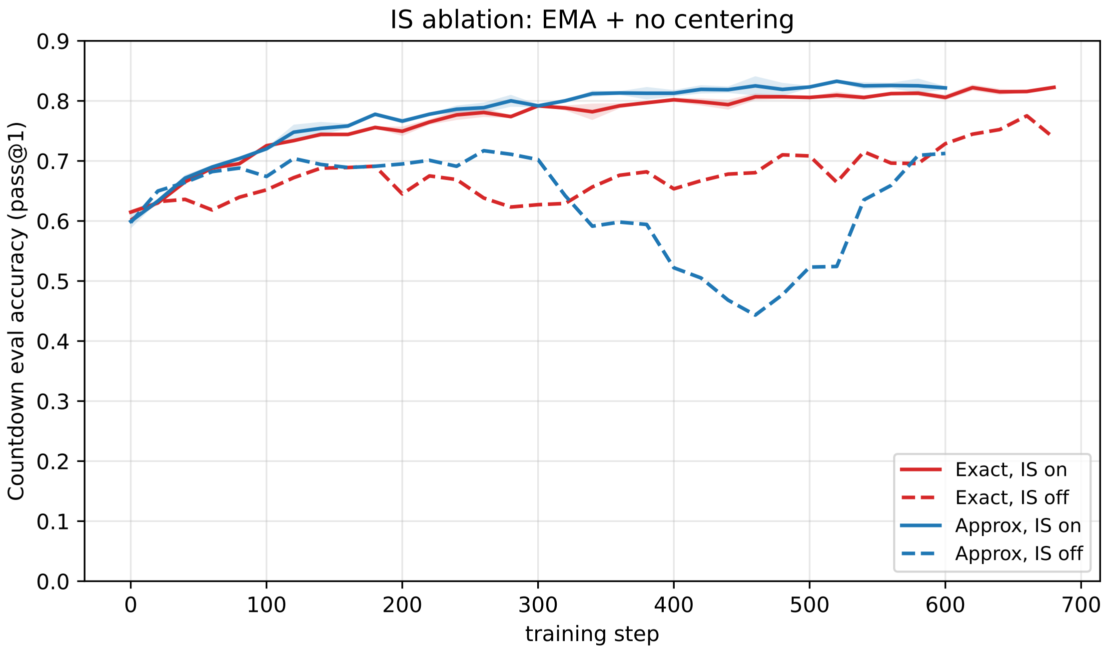

FIGURE 6. IS ablation on no-centering cells. Solid: IS on. Dashed: IS off. Red: exact KL (cell B). Blue: approx KL (cell D).

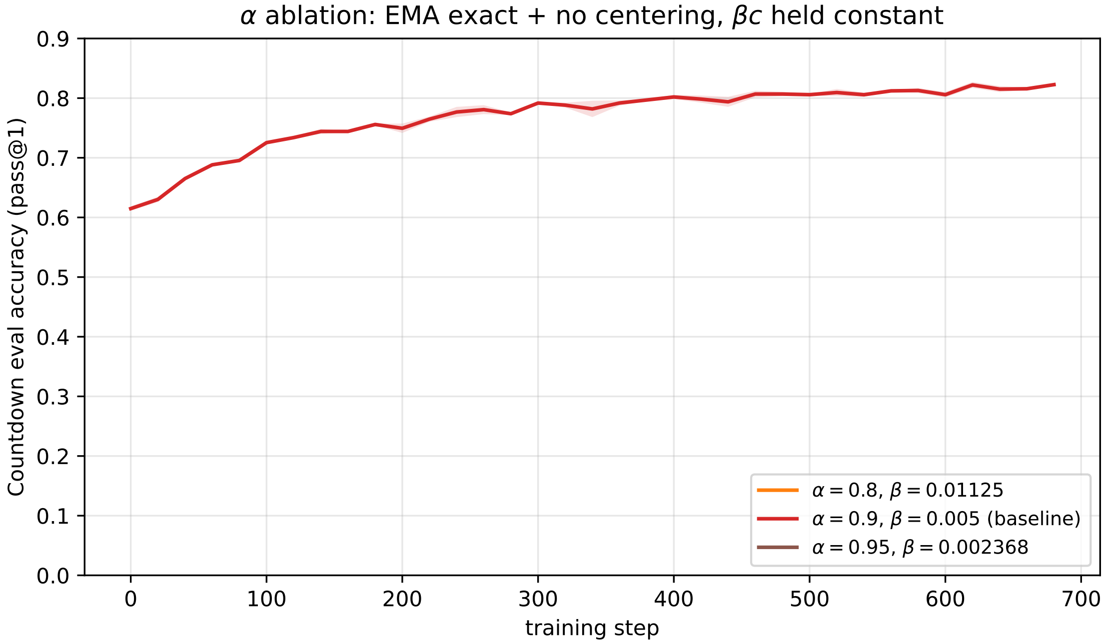

FIGURE 7. α ablation on no-centering exact KL with β scaled to keep $\beta \cdot c_\alpha$ constant.